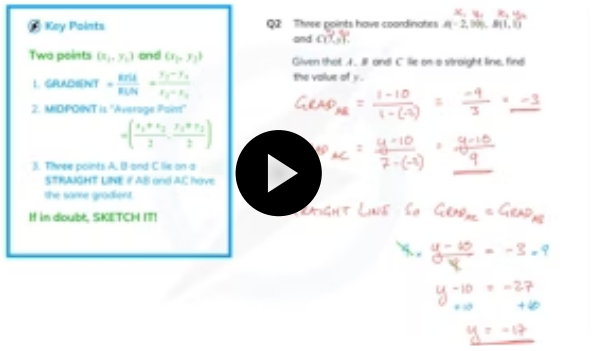
Gradient of a Line

Exam code: 0607

Download PDF

Written by: Mark Curtis
Reviewed by: Dan Finlay
Updated on 19 January 2026



Did this video help you? Yes No

Gradient of a line

What is the gradient of a line?

- The **gradient** is a measure of how **steep** a straight line is
- A gradient of 3 means:
  - For every **1 unit** to the right, go **up** by 3
- A gradient of -4 means:
  - For every **1 unit** to the right, go **down** by 4
- A gradient of 3 is **steeper** than 2
  - A gradient of -5 is **steeper** than -4
- A **positive** gradient means the line goes **upwards** (uphill)
  - Bottom left to top right
- A **negative** gradient means the line goes **downwards** (downhill)
- Top left to bottom right

How do I find the gradient of a line?

- Find **two points** on the line and draw a **right-angled triangle**
  - Then gradient $= \dfrac{\text{change in } y}{\text{change in } x}$
  - Or, in short, $\dfrac{\text{rise}}{\text{run}}$
    - The **rise** is the vertical length of the triangle

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

        - **Negative** for downhill lines
    - You can also find gradient of a line between two points, (x_1, y_1) and (x_2, y_2)
        - Use the formula $\frac{y_2 - y_1}{x_2 - x_1}$

How do I draw a line with a given gradient?

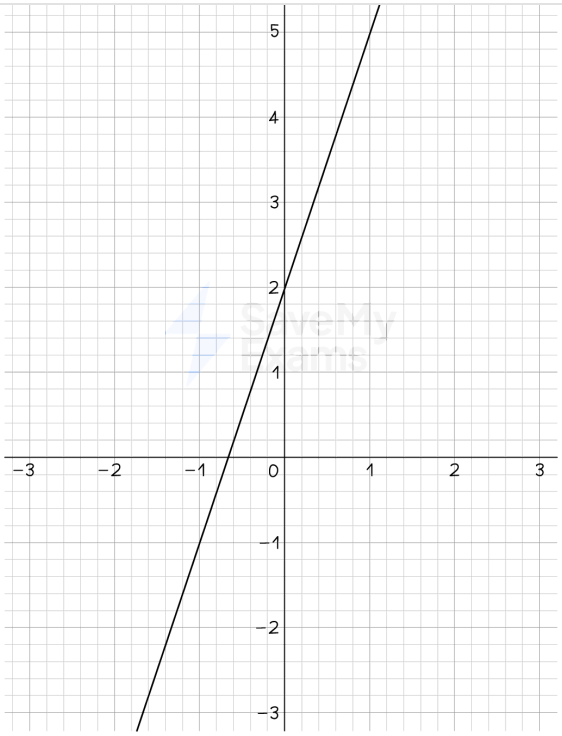
- To draw the gradient $\frac{2}{3}$
    - The **rise** is 2
    - The **run** is 3
    - It is **positive** (uphill)
        - Move 3 units to the **right** and 2 units **up**
- To draw the gradient -5 make it a fraction, $-\frac{5}{1}$
    - The **rise** is 5
    - The **run** is 1
    - It is **negative** (downhill)
        - Move 1 unit to the **right** and 5 units **down**

Examiner Tips and Tricks

A lot of students forget to make their gradients negative for downhill lines!

Worked Example

(a) Find the gradient of the line shown in the diagram below.



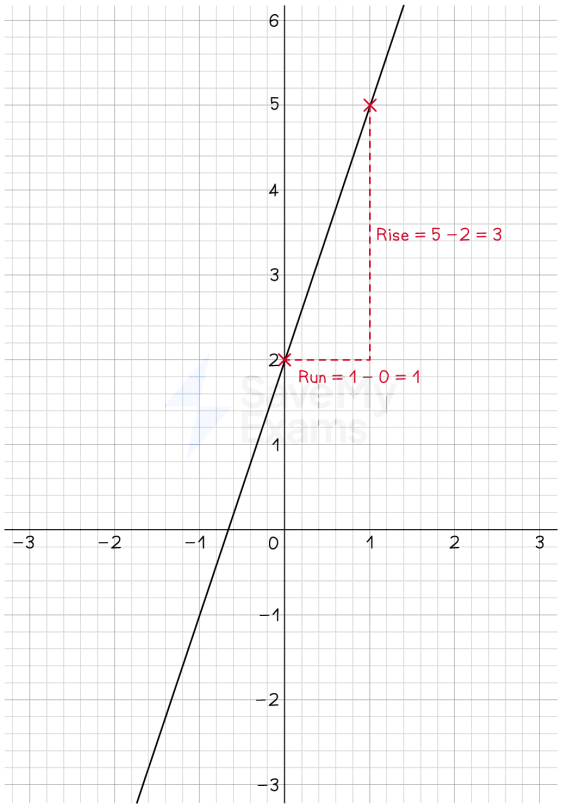
Copyright © Save My Exams. All Rights Reserved

Answer:

Find two points that the line passes through

$$(0, 2) \text{ and } (1, 5)$$

Use the grid to draw a right-angled triangle
Find the 'rise' (vertical length) and 'run' (horizontal length)



Copyright © Save My Exams. All Rights Reserved

$$\frac{3}{1} = 3$$

Look to see if the line is uphill or downhill

uphill, so the gradient is positive

The gradient is 3

(b) On the grid below, draw the line with a gradient of −2 that passes through (0,1).

Answer:

Mark on the point (0, 1)
-2 is the fraction $-\frac{2}{1}$
The rise is 2, the run is 1, the line goes downhill (so 1 across, 2 down)

(c) On the grid below, draw the line with a gradient of $\frac{2}{3}$ that passes through (0,-1).

Answer:

Mark on the point (0,-1)
The rise is 2, the run is 3, the line goes uphill (so 3 across, 2 up)

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub

Scholarship

Join

Dashboard

Overview

Brand ambassador

Profile

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.